

Photosynthesis and Cellular Respiration

Photosynthesis is the process by which green plants, algae, and some bacteria convert light energy into chemical energy.

The overall reaction combines carbon dioxide and water to produce glucose and oxygen, releasing oxygen as a byproduct.

Cellular respiration is the reverse process: mitochondria break down glucose using oxygen to release usable energy (ATP).

Together, photosynthesis and cellular respiration form a cycle that sustains most life on Earth by cycling carbon and energy.

Key vocabulary: chlorophyll, chloroplast, mitochondria, ATP, glucose, carbon cycle.